CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 17: write a prolog program to sum the integers from 1 to n

Aim:

To write and execute a Prolog program to calculate the sum of integers from 1 to N.

Algorithm:

1. Start the program.
2. Define the base condition: If N = 0, the sum is 0.
3. Check whether N is greater than 0.
4. Subtract 1 from N and store it in N1.
5. Recursively calculate the sum of integers from 1 to N1.
6. Add N to the previous sum.
7. Display the final sum.
8. Stop.

```
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

12 ?- emacs_consult_and_remove('C:/Users/navit/AppData/Local/Temp/xpce2').
true.

13 ?- sum(5, S).
S = 15
```

Result:

Thus, the Prolog program to calculate the sum of integers from 1 to N was successfully written and executed.